Phase 2: Design - Team 3 (Wednesday)

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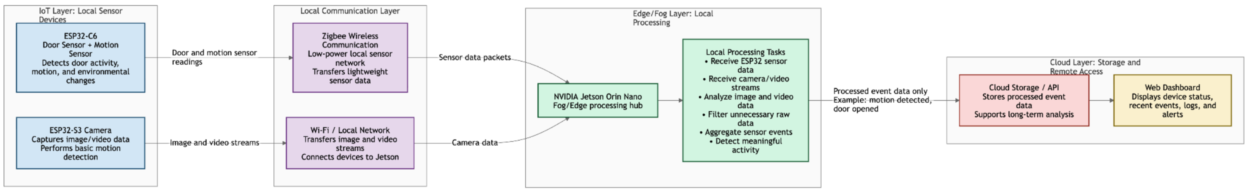
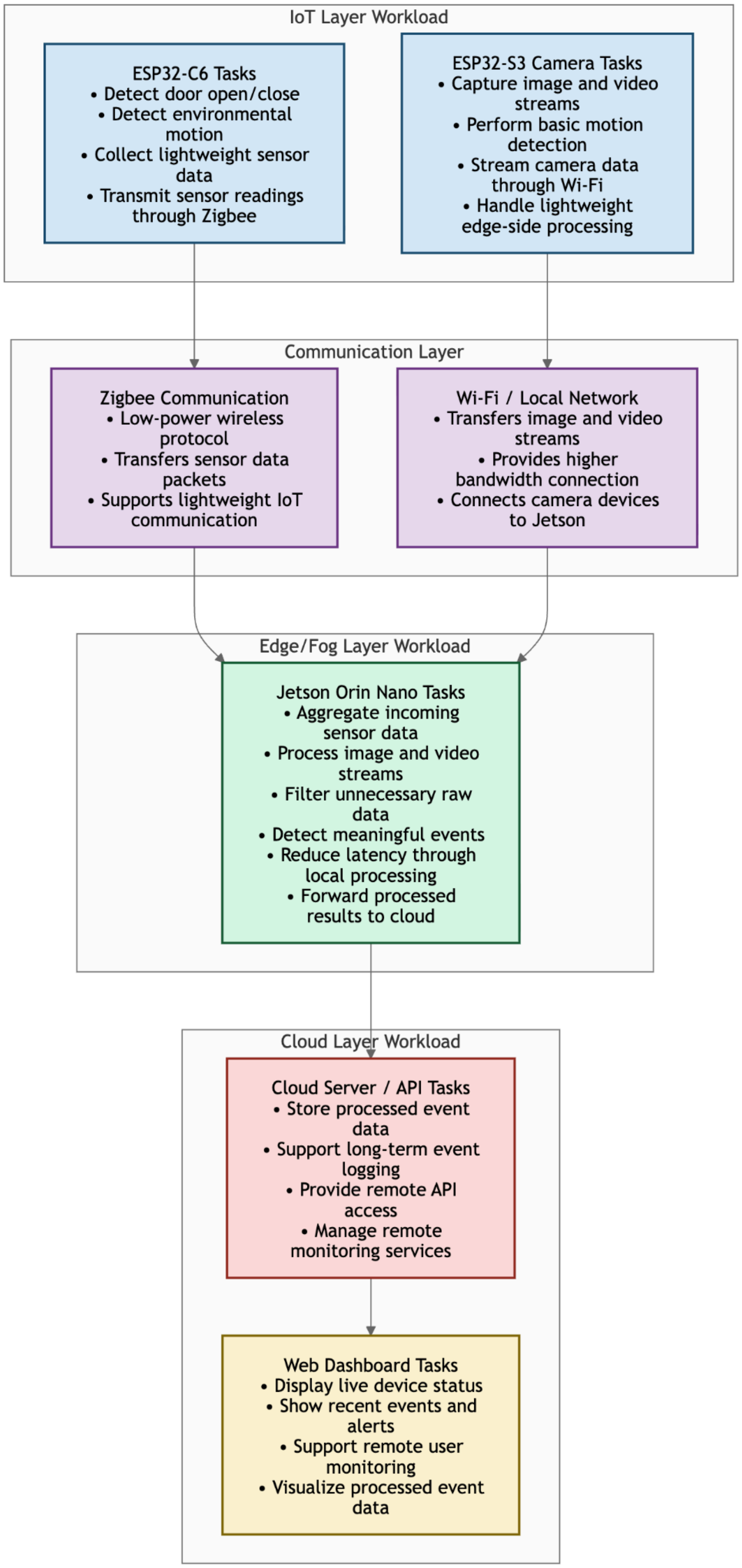
[Github]

1) System Design Diagram



The system consists of **4 main layers**: the **IoT layer**, **communication layer**, **edge/fog layer**, and **cloud layer**. The **ESP32-C6** and **ESP32-S3** collect sensor and video data, which is transmitted through **Zigbee** and **Wi-Fi** to the **NVIDIA Jetson Orin Nano** for local processing. Processed event information is then forwarded to **cloud storage and APIs**, while the **web dashboard** provides real-time remote monitoring and visualization.

2) Workload Distribution Diagram



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1. **IoT Layer (Data Collection)**

The **ESP32-C6** will handle lightweight sensing tasks such as detecting door activity, detecting environmental motion, collecting sensor readings, and transmitting small sensor packets through Zigbee. The **ESP32-S3** is responsible for capturing image and video streams as well as streaming camera data through Wi-Fi.

2. **Communication Layer**

The communication layer transfers information between devices. **Zigbee** is used for lightweight sensor communication because it is efficient for small sensor packets. Wi-Fi is used for image and video streams because camera data requires significantly more bandwidth and faster transfer speeds.

3. **Edge/Fog Layer**

The **Jetson Orin Nano** performs most of the computational workload in the system. It combines incoming sensor data, processes video streams locally, filters unnecessary raw data, and detects important events such as motion or door activity. By handling these tasks locally, the Jetson reduces latency and prevents large amounts of raw data from constantly being sent to the cloud.

4. **Cloud Layer**

The **cloud server and APIs** manage long-term storage and remote access. The cloud will store processed event information for future analysis. The **web dashboard** retrieves data to display live device status, alerts, and processed event data so users can remotely monitor the system in real time.

3) **A technical worksheet (tool inventory) of the devices, sensors, and other components of your proof of concept demo**

Device/Sensor/Component	Category	Main Task	Communication
NVIDIA Jetson Orin Nano	Edge Device	Process sensor and camera data locally	Wi-Fi and Local Network
ESP32-C6	IoT Sensor	Detects door activity and motion	Zigbee
ESP32-S3 Camera Module	IoT Camera	Captures image and video data	Wi-Fi
Motion Sensor (PIR) - **List which sensor**	Sensor	Detects environmental motion	Connected to ESP32-C6
Zigbee Network	Communication	Transfers sensor data	Zigbee
Wi-Fi Network	Communication	Transfers image and video streams	Wi-Fi
Cloud Storage/API	Cloud Infrastructure	Stores event data	Internet
Web Dashboard	User Interface	Displays alters and device status	Web Browser

4) **A report of how much each member of your team contributed (you can use a Scrum Sprint Report as a guideline for this)**

Member	Main Contributions	Current Tasks	Contribution
Marc Encarnacion	System planning	Designed initial system architecture	20%
Hikaru Shimada	IoT sensor setup	Assisted with ESP32-C6 setup	20%
Steven Vu	Camera module setup	Assisted with ESP32-C6 setup	20%
NJ Anandha Loganathan	Cloud/dashboard planning	Researched cloud storage and dashboard options	20%
Sujith Shen	Cloud/dashboard planning	Researched cloud storage and dashboard options	20%

5) **Showcase your repository with any updates on code you have written**

- [Github]